

Sprint 002 Academic Preprint: Uber H3 Spatial Indexing, Ring Generation, and Adjacency Mappings in the Web of Life

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Abstract

As global ecological and biogeochemical simulations scale, spatial discretization becomes a critical determinant of computational efficiency and thermodynamic fidelity. Sprint 002 of the **Web of Life** project establishes foundational geospatial infrastructure via Uber’s H3 hierarchical hexagonal spatial index. Implemented within `src/spatial/h3_adjacency.ts`, this sprint introduces rigorous H3 string parsing, K -ring neighborhood generation, and edge-neighbor mapping routines. By treating individual H3 cells as open thermodynamic control volumes (Ω_i), we formulate mass and energy flux transfers that strictly adhere to the **First Law of Thermodynamics** ($\sum \Delta M = 0$). Monadic state wrappers (`SpatialMonad<T>`) further ensure side-effect-free, composable ecological state transformations across spatial boundaries. The official repository is maintained at <https://github.com/pascalranoroarijaona/WebOfLife>.

1 Introduction and Thermodynamic Context

The Web of Life simulation engine models planetary-scale trophic energy transformations, carbon cycling, and biogeochemical flows. To avoid the topological distortions inherent in traditional rectangular grids (such as diagonal distance inequities), Sprint 002 integrates Uber’s H3 hierarchical hexagonal spatial index.

Each control volume Ω_i operates as an open thermodynamic system capable of exchanging mass (carbon, water, mineral nutrients) and thermal-radiant energy with its immediate hexagonal edge neighbors $j \in \text{Adj}(i)$.

In accordance with the **First Law of Thermodynamics**, the net change of any conserved stock S within control volume Ω_i over time interval Δt is governed by internal biological/abiochemical sources/sinks G_i and boundary fluxes across the six hexagonal edges (e_{ij}):

$$\frac{dM_i}{dt} = G_i + \sum_{j \in \text{Adj}(i)} J_{ji} \quad (1)$$

where J_{ji} represents the directed mass or energy flux vector from adjacent cell j to cell i .

2 Mathematical Formalization of H3 Topology

2.1 Cell Resolution & Area Scaling

H3 resolutions $r \in [0, 15]$ dictate the surface area $A(r)$ and characteristic edge length $L(r)$ of each control volume. For biological and geochemical tracking, standard resolutions (e.g., $r = 3$ to $r = 5$) provide optimal mesoscale control volumes:

- K -ring size cardinality for a ring of radius k follows the exact hexagonal scaling law:

$$N_{\text{cells}}(k) = 3k^2 + 3k + 1 \quad (2)$$

- Edge neighbor count for standard hexagonal cells is strictly invariant:

$$|\text{Adj}(i)| = 6 \quad (3)$$

3 Architectural Design & Implementation

To maintain modularity and architectural purity, Sprint 002 introduces a dedicated spatial namespace and class hierarchy within `src/spatial/h3_adjacency.ts`.

3.1 Interface Contracts

```
export interface IH3SpatialCell {
  readonly index: string;
  readonly resolution: number;
  readonly baseCell: number;
  getEdgeNeighbors(): string[];
  getKRing(k: number): string[];
}

export interface IH3AdjacencyEngine {
  parseIndex(h3Str: string): IH3SpatialCell;
  generateKRing(center: IH3SpatialCell, k: number): string[][];
  getEdgeNeighbors(cell: IH3SpatialCell): Map<number, string>;
}
```

3.2 Spatial Monads and Diffusion Fluxes

Spatial states are encapsulated within monadic containers to facilitate functional composition:

$$\mathbf{M}(\text{CellState}) \xrightarrow{\text{H3Adjacency}} \mathbf{M}(\text{NeighborhoodFlux}) \quad (4)$$

Fickian diffusion across hexagonal boundaries computes conservative stock adjustments for carbon (C), water (H_2O), and thermal energy (E_{th}), ensuring that system-wide residuals satisfy:

$$\Delta M_{\text{system}} = \Delta M_{\text{center}} + \sum_{j \in \text{Adj}(i)} \Delta M_j = 0 \quad (\pm \epsilon_{\text{machine}}) \quad (5)$$

4 Conclusion and Future Horizons

Sprint 002 successfully bridges abstract ecological modeling with rigorous spatial indexing. By establishing invariant hexagonal topologies and conservative mass-energy diffusion operators, the Web of Life engine is fully equipped for multi-resolution biome simulations.

For further details, implementation code, and test suites, consult the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.